AnchorMTM: Spatially Anchored Merge Tree Maps for Time-Varying Scalar Fields

Category: Research

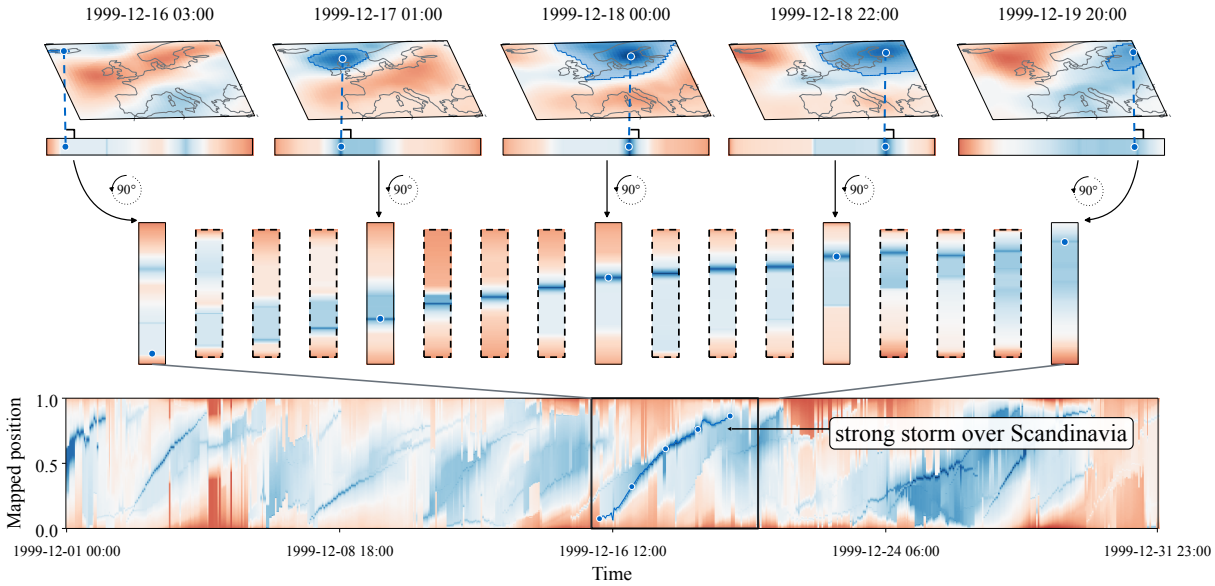


Figure 1: Storms example linking spatial feature supports to a temporal map ($\varepsilon = 5\%$). Top: five snapshots of one matched extremum and their scalar profiles; blue outlines mark its leaf-arc supports, and dashed guides link the extrema to their displayed profile anchors. Middle: the profiles rotated by 90°; dashed borders distinguish sampled intermediate frames. Bottom: the full 744-frame map, with the selected storm-related region outlined and the matched feature's displayed positions marked in blue. The vertical map axis is the normalized reference coordinate, not longitude.

Abstract

Temporal merge tree maps summarize evolving scalar topology in one image, but their display axis need not correspond to any spatial direction. AnchorMTM anchors the display axis to a fixed spatial reference, estimated once from matched extremum motion or supplied, and exposes one control, the merge-error budget ε: Exact Mode keeps the merge tree, and Budgeted Mode trades a declared merge-level error for position. A topology-budgeted order search with optimal merge barriers and a capped quadratic program construct the map. Three certified bounds state the displacement that exact topology, finite widths and bounded topological error make unavoidable; they depend only on the layout constraints, not on the algorithm. On seven planar and volumetric sequences (917 frames), Exact Mode comes within 0.03 percentage points of the exact-topology bound on every planar sequence with an unchanged merge tree, whereas 31–95% of the order error of TMTM and ST-MTM lies above that bound and could be avoided without changing the tree; and ε traces the position–topology trade-off with every rendered map within its allowance.

Index Terms: Time-varying scalar fields, merge tree maps, spatial anchoring, topology preservation, constrained layout.

1 Introduction

2D and 3D time-varying scalar fields describe evolving phenomena such as pressure systems, combustion structures, and flow features. Their analysis requires understanding where features occur, how they move along a relevant spatial direction, and how their hierarchical relationships change throughout a sequence. Extrema and component merges describe this evolving scalar structure, while spatial coordinates connect it to the physical domain. A temporal overview should bring these aspects together. Figure 1 illustrates how spatial supports and reference positions connect to scalar profiles and the resulting temporal map.

Animation preserves the spatial view at each instant, but comparison between distant times requires navigation and memory. Small multiples expose observations simultaneously at the cost of screen space. Static temporal summaries allocate one display axis to time and compress the spatial domain into the remaining axis [1,31,34]. Spatial linearization makes long sequences accessible in one image; the meaning of its 1D coordinate determines which spatial questions the overview can support.

Merge tree maps organize this compression around scalar features and their hierarchy. Temporal Merge Tree Maps (TMTM) linearize augmented merge trees into scalar profiles aligned across time [22]. Minima and intervening barriers encode extrema and merge levels, exposing scalar structure and its temporal evolution. Spatiotemporal Merge Tree Maps (ST-MTM) incorporate relative spatial distances and temporal position constraints into a geometry-aware layout [9]. These methods provide the foundation for compact overviews of temporal scalar topology.

A display coordinate chosen for hierarchy, relative distance, or temporal alignment need not correspond to a fixed physical direction. Its position cannot then be read directly as longitude or distance along a flow axis, and its changes need not represent motion along that direction. Relative geometry supplies spatial relationships but leaves the absolute coordinate origin undetermined. The remaining question is how to give the temporal map a consistent spatial reference while retaining its hierarchical information.

We introduce *Automatic Spatial Anchoring*: estimating one dom-

inant motion direction from matched extrema over the full sequence, assigning projected targets in that shared coordinate system, and placing display anchors nearby as constraints permit. A task-specific physical direction remains an option. The direction estimator is a standard leading-eigenvector calculation; our contribution is its connection to merge-tree layout, explicit topology budgets and position bounds. A reference order can conflict with topology when the projected leaves of different subtrees interleave while exact topology requires contiguous subtree blocks. Finite feature widths introduce further packing constraints. Allowing bounded topological error releases some ordering restrictions and exposes a controlled position–topology trade-off.

AnchorMTM, *Spatially Anchored Merge Tree Maps*, follows one construction: extract and match features, construct a fixed reference, seek a topology-admissible leaf order, optimize anchors within legal intervals, and reconstruct scalar profiles on a shared temporal axis. Exact Mode and Budgeted Mode use the same layout framework with different admissible merge errors. Position bounds then explain the discrepancy that topology and finite widths make unavoidable.

Our main contributions are as follows:

- We formulate spatially anchored merge tree maps as hierarchy-constrained interval layouts with a fixed reference, a declared merge-level allowance and a position cap, and identify how a spatial order conflicts with the merge tree.
- We introduce AnchorMTM, which gives the display axis a spatial meaning through a fixed reference, topology-budgeted leaf ordering with optimal merge barriers and capped anchor optimization, governed by one merge-error budget.
- We establish three certified position bounds that hold for every layout meeting their constraints, and show on seven sequences that Exact Mode nearly attains them on planar data while most of the order error of TMTM and ST-MTM lies above them.

2 Related Work

2.1 Spatially Referenced Temporal Visualization

Static temporal displays expose distant observations simultaneously, while animation retains the spatial view of each instant; their usefulness depends on the analytical task [31, 34]. Generalized space–time cubes describe the transformations connecting such views [1]. In compressed temporal overviews, the spatial coordinate determines which location and motion relationships remain directly interpretable.

Hovmöller diagrams retain a selected physical coordinate and summarize the other dimensions [20]. Geographic orderings and space-filling curves instead traverse the domain [8, 16]. Dynamic Volume Lines and data-driven curves extend spatial traversal to volumetric comparison and scalar-aware ordering [37, 39]. A traversal can retain spatial neighborhoods, but its 1D coordinate is generally different from a fixed projection, and traversal adjacency alone does not enforce the merge hierarchy.

1D projections organize spatiotemporal phenomena [14], and MotionRugs expose collective behavior of moving entities [5]. More generally, an embedding trades original-domain relationships against its display organization [28]. Relative proximity does not determine an absolute coordinate. AnchorMTM assigns fixed projection targets to topological features and quantifies their displacement under hierarchical and interval constraints, combining spatial reference with a temporal topology overview.

2.2 Topology-Based Temporal Maps and Geometry-Aware Layouts

Topology-based visualization organizes scalar structures through connectivity [17]. Landscapes and landscape profiles express hierarchies through terrain or 1D scalar representations [29, 35], with geometry-preserving and temporally coherent extensions [3, 18]. These representations connect scalar values and feature relationships in a compact view.

TMTM emphasizes intact topological features and temporal alignment in static merge-tree maps [22]. ST-MTM incorporates relative spatial geometry and temporal position constraints [9]. These priorities give spatial relationships and coherence a role in the map, while leaving its display coordinate free to differ from a fixed physical direction. AnchorMTM estimates a shared direction from extremum motion, or accepts a physical direction, and couples that reference to topology-budgeted layout and certified position bounds. It reuses the established profile and scalar-filling foundation.

Hierarchy-constrained leaf ordering traditionally minimizes adjacent dissimilarity [2]; target-order reordering studies related tree-consistent permutations [6]. Separation-constrained layouts control geometric overlap and displacement [11,24]. AnchorMTM couples these layout concerns to fixed spatial targets. Its exact-topology point-order bound has a closed form, whereas finite-width candidate ordering remains heuristic.

Merge-tree comparison depends on the descriptors and correspondences used [38]. Labelled cophenetic/interleaving comparisons support fixed-identity merge-level accounting [15, 27], and ordered merge trees connect scalar curves to monotone interleaving distances [4]. We use fixed labels to measure displayed merge error. The optimal-barrier subproblem relates to minimax ultrametric fitting [13], with the additional restriction that one linear order induces every displayed merge.

3 Background

3.1 Augmented Merge Trees and Feature Supports

For a sampled 2D or 3D field $f_t : D \to \mathbb{R}$, the merge tree T_t records sublevel-set connectivity: minima create components and saddles join them as the threshold increases. Negating the field gives the split-tree convention. Critical points are supernodes connected by superarcs; augmentation associates regular vertices with these arcs [9, 22, 32]. Extraction and simplification precede layout.

A leaf represents an extremum at spatial location p_i^t. Its feature support S_i^t consists of that extremum and the regular samples on its incident leaf arc, excluding the parent critical sample. It is not the full component or the complete subtree above that arc. For cross-frame overlap, these supports are sets of stable spatial sample IDs: the same ID must denote the same spatial sample across frames. Fixed-grid inputs preserve this identity through consistent sample indexing. Internal arcs retain further regular samples for scalar filling. Write $\mathscr{D}_t(v)$ for the descendant leaf labels of a node v, and $h_t(v)$ for its scalar level. The lowest common ancestor determines the merge-height matrix:

$$V_{ij}^t = h_t(\mathrm{lca}(i,j)), \quad i \neq j; \qquad V_{ii}^t = f_t(p_i^t). \tag{1}$$

Equal-level merge events form one multifurcation; arbitrary binary refinements do not add strict constraints. All subsequent guarantees refer to the supplied, possibly simplified tree.

3.2 One-Dimensional Merge-Tree Profiles

A 1D merge-tree profile represents extrema at anchors in a selected leaf order. Regular vertices on leaf arcs supply scalar context within leaf intervals; internal arcs and paths to the LCA supply the intervening gaps [9, 22]. In join convention, extrema become profile minima, and the largest profile value between two anchors gives their displayed merge level. Exact preservation requires each strict subtree to occupy a contiguous leaf block. Arranging these profiles along time yields a static temporal map. This profile model provides the basis for the spatial anchoring formulation below.

4 AnchorMTM

We first state the layout model, the admissible maps and the fidelity measures (Section 4.1). AnchorMTM then fixes a spatial reference (Section 4.2) and constructs the map with one control (Section 4.3); certified bounds state the displacement that no layout in the model can avoid (Section 4.4). Analysis uses join convention; split-tree scalar signs are restored after reconstruction.

4.1 Layout Model and Fidelity Measures

A merge tree map chooses, per frame, a leaf order, extremum anchors, and a scalar profile:

$$M_t = (\pi_t, x_t, \psi_t), \qquad \psi_t : [0,L] \to \mathbb{R}. \tag{2}$$

Leaf i has a reference position $q_i \in [0,L]$ on the display axis (Section 4.2 constructs it) and a width $W_i^t = s\,m_i^t$ from its positive support measure with a fixed scale s, without per-frame normalization. The interval-layout set $\mathscr{L}(\pi)$ requires intervals of widths W_i centered at z_i to lie inside the canvas, remain disjoint with gap at least $g \geq 0$, and contain anchors satisfying $|x_i - z_i| \leq \rho W_i/2$, $0 \leq \rho \leq 1$. Infeasible total widths and gaps are reported rather than rescaled.

Actual displacement is $|x_i - q_i|$, normalized by L when reported. To distinguish order conflict from finite-width placement, define

$$\mathrm{NROE}(\pi) = \frac{1}{2L} \max_{j \leq k} [q_{\pi(j)} - q_{\pi(k)}]^+. \tag{3}$$

This is the smallest worst displacement of a nondecreasing point placement, divided by L; it ignores widths. For finite intervals, the exact fixed-order cost is

$$\tau(\pi) = \min_{(x,z) \in \mathscr{L}(\pi)} \|x - q\|_\infty. \tag{4}$$

The minimum is attained whenever the layout set is nonempty. It is evaluated exactly as a minimax linear program.

For distinct labels, let V_{ij}^{ψ} be the maximum profile value between their anchors, with $V_{ii}^{\psi} = \psi(x_i)$. A map admits merge-level allowance $\eta \geq 0$ if

$$d_{\mathrm{top}}(\psi_t, T_t) = \max_{i,j} |V_{ij}^{\psi_t} - V_{ij}^t| \leq \eta. \tag{5}$$

AnchorMTM sets $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, where $\Delta_F > 0$ is the sequence-wide scalar range, ε_0 is a fixed numerical allowance, and $\varepsilon \geq 0$ is the sole primary user control. The implementation defaults to $\varepsilon_0 = 10^{-7}$; it only resolves saddles tied to within $\varepsilon_0 \Delta_F$. NMLE reports the achieved normalized error $d_{\mathrm{top}}/\Delta_F$. Strict exact topology is $\eta = 0$: zero labelled error on the declared tree.

For canonical cophenetic merge-tree representations whose fixed labels cover every leaf and whose induced matrices include the diagonal, d_{top} equals the unnormalized labelled interleaving distance [15, 27]. Under these conditions, $d_B \leq d_I \leq d_{\mathrm{top}}$ [26].

A map is *admissible* for allowance η if its intervals form a layout in $\mathscr{L}(\pi_t)$ with anchors x_t, its profile shows every extremum at its value, $\psi_t(x_i^t) = V_{ii}^t$, and $d_{\mathrm{top}}(\psi_t, T_t) \leq \eta$. Let $\Pi_\eta(T_t)$ contain the orders that admit such a profile; Proposition 1 gives the exact test $\delta_t^*(\pi) \leq \eta$. For $\eta = 0$, every strict subtree occupies a contiguous block. The bounds of Section 4.4 concern admissible maps only, whatever their objective or solver; AnchorMTM chooses among them by a fixed reference, a position cap and the objective of Eq. 11.

4.2 Spatial Reference

Cross-frame Leaf Correspondence. We track merge-tree leaf extrema p_i^t using their leaf-arc supports S_i^t. On a shared fixed grid, overlap-based preprocessing [9] counts common spatial samples:

$$O_{ij}^t = |S_i^t \cap S_j^{t+1}|. \tag{6}$$

Hungarian assignment maximizes total intersection count over one-to-one pairs, discarding zero-overlap pairs.

Continuation inherits a Track ID; *birth/death* starts an unmatched current identity or ends an unmatched previous track. Only one-to-one links are retained; identity states do not identify physical split or merge events. For a current leaf i, $\mu_t(i)$ denotes its valid predecessor.

These correspondences, computed before any topology-budget sweep, provide the motion samples for Spatial Anchoring, which distinguishes a reference target q_i^t from the display anchor x_i^t. By default, one unit direction $a^* \in \mathbb{R}^d$, $d \in \{2,3\}$, is estimated offline from all matched consecutive-frame extremum displacements:

$$\begin{aligned} \Delta p_i^t &= p_i^t - p_{\mu_t(i)}^{t-1}, \\ S &= \sum_{t,i \text{ matched}} \Delta p_i^t (\Delta p_i^t)^\top, \\ a^* &\in \arg\max_{\|a\|=1} a^\top S a. \end{aligned} \tag{7}$$

The leading eigenvector maximizes squared projected motion, not feature count or topological compatibility. The same calculation serves 2D and 3D; it is performed once for the whole sequence. Sign, degenerate and tied cases follow fixed rules given in the supplement. A supplied physical direction may replace a^* without changing the layout. Overlap matching may miss fast-moving features or those with changing or weakly overlapping supports, affecting the displacement sample and hence the estimated direction.

For the selected direction, the minimum and span of the projected domain bounding-box corners define a common origin o and extent $L > 0$. The reference is

$$q_i^t = \langle a^*, p_i^t \rangle - o \in [0,L]. \tag{8}$$

Direction, origin and scale remain fixed across frames and all budgets for the same prepared sequence. Distinct extrema can share projected coordinates while retaining separate identities. Anchoring minimizes constrained deviation rather than requiring $x_i^t = q_i^t$.

Relative geometry uses distances d_{ij}^t between leaf-support centroids. These differ from extremum references because a changing support can move its centroid while the extremum stays fixed.

4.3 Spatially Anchored Layout Construction

All remaining settings belong to a fixed configuration Θ_0: position slack β_0, the four objective weights, width scale, gap, anchor slack and output resolution. For a prepared sequence, the default interface is $M_t^\varepsilon = \mathrm{AnchorMTM}(T_t, q_t, W_t, V_t, \mu_t; \varepsilon, \Theta_0)$. Advanced settings may be supplied once, but only ε varies in a budget sweep. Supplementary Material specifies the fixed scale rules and implementation tolerances.

With features and their reference fixed, construction chooses an order by the lexicographic objective

$$\begin{aligned} \pi_t^\varepsilon \in \arg\min_{\pi \in \mathscr{C}_t(\varepsilon)} & \big(\tau_t(\pi), \mathrm{NROE}_t(\pi)\big), \\ \text{s.t.} \quad & \delta_t^*(\pi) \leq \eta(\varepsilon). \end{aligned} \tag{9}$$

The finite candidate pool contains the original strict-topology order, midpoint initializations, and orders visited during width-aware refinement of threshold and selective flattenings. Selection is not a global optimum over all permutations.

4.3.1 Topology-Budgeted Leaf Ordering

Exact Mode sets $\varepsilon = 0$, leaving only the numerical allowance $\eta = \varepsilon_0 \Delta_F$; on every planar sequence its rendered merge error is zero. *Budgeted Mode* permits $\varepsilon > 0$. Both modes share the same interval model and placement objective.

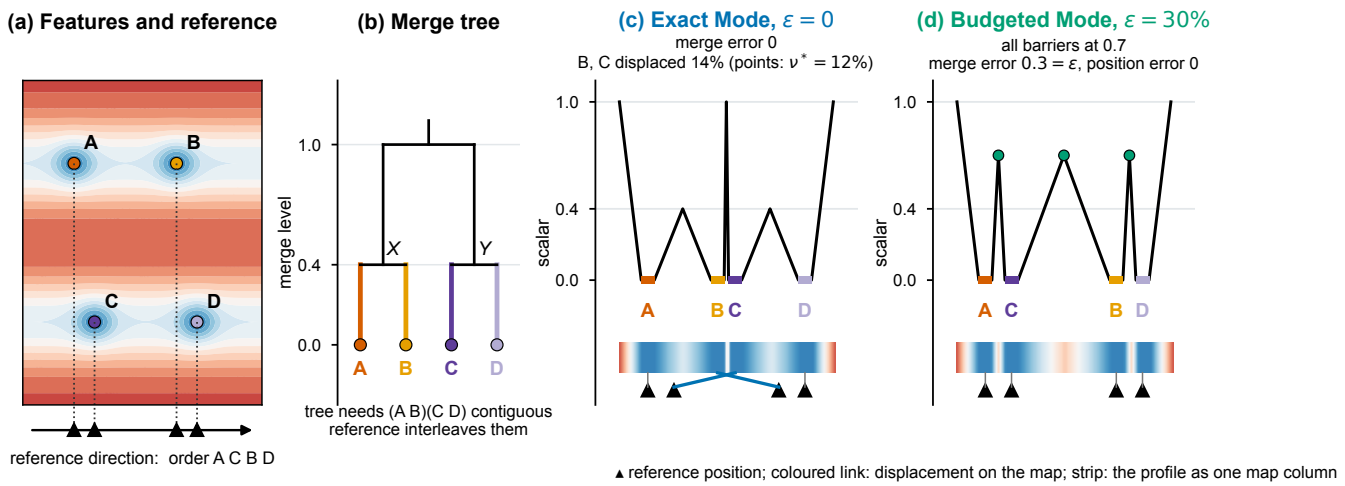


Figure 2: Why anchoring conflicts with the hierarchy, and what the allowance buys, on the four-leaf example of Section 5.2. (a) Projected onto the reference, the minima are ordered A C B D; (b) the merge tree requires the sibling blocks (A B) and (C D). (c) Exact Mode keeps the blocks contiguous and displaces B and C by 14% (12% for points, Theorem 1). (d) With an allowance of half the parent gap, $\varepsilon = 30\%$, the reference order is admissible: all three barriers are set to 0.7, so every displayed merge level is off by exactly $0.3 = \varepsilon$, and no feature is displaced. Each strip below a profile is the corresponding column of the temporal map.

Algorithm 1: Single-Control AnchorMTM

Data: 2D/3D field sequence; $\varepsilon \geq 0$; fixed configuration Θ_0.
Result: Temporal map I^ε and error records, or a failure report.
Extract trees, supports and extrema; match supports; estimate one direction and references q_t (once)
$\eta \leftarrow (\varepsilon_0 + \varepsilon)\Delta_F$
for *each time step* t **do**
  $\mathscr{F}_t \leftarrow \{\pi \in \text{GenerateAndRefine}(T_t, q_t, W_t, V_t, \eta) : \delta^*(\pi) \leq \eta\}$; report if empty
  Select $\pi_t^\varepsilon \in \mathscr{F}_t$ by (τ, NROE); set optimal merge barriers
  Solve the QP for π_t^ε with cap $\tau(\pi_t^\varepsilon) + \beta_0$; reconstruct the profile; record errors
return I^ε *and records*

Candidate generation includes the exact tree, threshold flattenings, and a conflict-driven selective flattening. Midpoint orders initialize these structures. Width-aware block insertion refines the four best initial structures, together with the exact tree, while retaining the point-model budget order as a candidate. Moves improve the lexicographic score consisting of τ and a squared packing-residual term, and must pass $\delta^*(\pi) \leq \eta$.

Guards report capacity failures, empty admitted sets and solver failures instead of silently rescaling widths, gaps or canvas; the supplement details them.

4.3.2 Optimal Merge Barriers and Topological Feasibility

For a fixed order, a pair's displayed merge is the largest intervening barrier. Set $C_k = \min_{i \leq k < j} V_{\pi(i)\pi(j)}$ and $F_k = \max(V_{\pi(k)\pi(k)}, V_{\pi(k+1)\pi(k+1)})$ for gap k.

Proposition 1 (Optimal barriers)**.** *With fixed extremum values in the continuous non-strict model, the smallest labelled merge error is*

$$\delta^*(\pi) = \max\left\{0, \max_k (F_k - C_k), \frac{1}{2}\max_{i<j}\left(V_{\pi(i)\pi(j)} - \max_{i \leq k < j} C_k\right)\right\}, \tag{10}$$

and barriers $b_k = C_k + \delta^*(\pi)$ *attain it.*

Error δ requires $F_k \leq b_k \leq C_k + \delta$. Choosing the largest admissible barriers satisfies every pair's lower merge requirement exactly when $\delta \geq \delta^*$. Thus the order meets the merge-level allowance in this continuous model exactly when $\delta^*(\pi) \leq \eta$.

4.3.3 Spatially Anchored Spatiotemporal Placement

Once the order is fixed, the core terms are E_R, mean squared deviation from the fixed reference, and E_M, mean squared error in matched projected motion. The auxiliary term E_G approximates support-centroid distances through ordered anchor separations; E_C regularizes anchor offsets from interval centers. All four terms and their nonnegative weights are retained:

$$\begin{aligned} \min_{x_t, z_t} \quad & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t.} \quad & (x_t, z_t) \in \mathscr{L}(\pi_t), \\ & \|x_t - q_t\|_\infty \leq \tau(\pi_t) + \beta_0. \end{aligned} \tag{11}$$

For each valid match, set $\Delta q_i^t = q_i^t - q_{\mu_t(i)}^{t-1}$ and use the previous display anchor plus this reference displacement as the temporal target:

$$\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1}). \tag{12}$$

Thus E_M follows projected motion while favoring continuity of $x - q$. First-frame, newly born and otherwise unmatched leaves have no previous-frame motion penalty.

Only ε changes the topology allowance $\eta(\varepsilon)$. The fixed slack $\beta_0 \geq 0$ bounds additional position deviation after ordering. The cap is feasible because $\tau(\pi_t)$ is attained, and the QP cannot change the selected order. The fixed-order problem is convex; sequential processing does not imply a joint sequence optimum or exact preservation of every projected motion.

4.3.4 Profile Reconstruction and Output Diagnostics

Reconstruction inherits the leaf-arc sampling and LCA-path filling of the merge tree map pipeline [9, 22]. Original extrema remain at their anchors; regular samples provide scalar context within intervals and gaps. Optimal-barrier correction then clips each inter-anchor interior to b_k and raises an existing peak if necessary.

A common direction, origin, extent, and pixel conversion are used for all frames. The profiles form the final temporal map,

$$I^{\varepsilon}(t,u) = \psi_t^{\varepsilon}(u), \qquad 0 \le u \le L. \tag{13}$$

Resampling, strict-extremum clearance and raster rounding can change the achieved error, so records report rendered errors separately from δ^*.

4.4 Certified Position–Topology Bounds

The construction can leave reference displacement even after optimization. The following bounds explain what topology and finite widths make unavoidable: the first answers how much a tree-compatible point order must deviate, the second adds interval packing, and the third asks which restrictions survive a merge-error allowance. They are benchmarks, not pruning rules; full proofs are in Supplement S2–S5. We suppress t within a frame, use $[v]^+ = \max(v,0)$, and take empty maxima as zero.

4.4.1 Exact-Topology Reference-Order Bound

For a reduced tree with strictly increasing proper-ancestor levels, exact topology makes every subtree contiguous. Let $[m_X, M_X]$ be the reference hull of the leaves in subtree X.

Theorem 1 (Exact-Topology Position Bound)**.** *Every zero-error profile has* $\mathrm{NROE}(\pi) \ge \nu^*(T,q)$*, where*

$$\nu^*(T,q) = \frac{1}{2L} \max_{v} \max_{\substack{X,Y \in \mathrm{ch}(v) \\ X \ne Y}} [\min(M_X - m_Y, M_Y - m_X)]^+. \tag{14}$$

Recursive child ordering by reference-hull midpoint attains this point-model order bound in $O(n \log n)$ time.

Proof sketch. Sibling blocks must choose one orientation. Their two possible inversion costs are $[M_X - m_Y]^+/2$ and $[M_Y - m_X]^+/2$. Midpoint order chooses the smaller; every inverted leaf pair is separated by sibling blocks at its LCA. □

This result isolates the ordering restriction imposed by the tree and references. Midpoint ordering attains the point-order optimum; finite intervals introduce additional constraints, considered next.

4.4.2 Width-Aware Position Bound

Finite widths, minimum gaps, anchor slack and canvas capacity add packing costs to the point-order obstruction. For a nonempty leaf set S, let $H_i = \rho W_i/2$, $\ell_i = q_i - W_i/2 - H_i$, $h_i = q_i + W_i/2 + H_i$, and $s_i = W_i + g$. Define two packing summaries:

$$\begin{aligned} A(S) &= \max_{a \in S} \left(\ell_a + \sum_{\substack{i \in S \\ \ell_i \ge \ell_a}} s_i \right), \\ B(S) &= \max_{b \in S} \left(\sum_{\substack{i \in S \\ h_i \le h_b}} s_i - h_b \right). \end{aligned} \tag{15}$$

The first uses upper reference-endpoint sets and the second lower sets; sorting and cumulative sums evaluate them. For disjoint leaf sets X, Y and the full leaf set $\mathscr{F}$, write

$$\begin{aligned} \omega(X,Y) &= \tfrac{1}{2} \min\{A(X) + B(Y), A(Y) + B(X)\} - \tfrac{g}{2}, \\ E_\partial &= \max\{0, B(\mathscr{F}) - g, A(\mathscr{F}) - g - L\}. \end{aligned} \tag{16}$$

Here ω takes the cheaper block orientation and E_∂ accounts for both canvas ends. Let $\mathscr{P}_0$ contain every pair of distinct sibling descendant-leaf sets. The certificate is

$$\mathscr{B}_w(T,q,W) = \max\{E_\partial, \max_{(X,Y) \in \mathscr{P}_0} \omega(X,Y)\}. \tag{17}$$

The dependence on g, ρ and L is through Eqs. 15–16; it is suppressed in the certificate's arguments.

Theorem 2 (Width-Aware Position Bound)**.** *For any feasible exact-topology interval layout,*

$$\|x - q\|_\infty \ge \tau(\pi) \ge \mathscr{B}_w(T,q,W). \tag{18}$$

With zero widths and gaps, $\mathscr{B}_w = L\nu^*(T,q)$.

Proof sketch. The first and last anchors of selected leaves must enclose their intervals and intervening gaps, up to the allowed anchor slack. Upper and lower sets in Eq. 15 maximize the resulting shortfall against reference separation. Exact topology requires one orientation of every sibling pair; the canvas ends give the two one-sided costs. □

Thus $\mathscr{B}_w$ bounds all exact-topology legal interval orders and need not be attained; $\tau(\pi)$ is the attained minimum for a specified order. These quantities have coordinate units, whereas NROE and ν^* are normalized by L.

4.4.3 Bounded-Topology Position Bound

A merge-level allowance can release some contiguous blocks, but still limits interleaving. If leaf k lies between descendants i, j of node u, interval containment gives $V_{ik}^{\psi} \le V_{ij}^{\psi}$, hence $V_{ik} \le h(u) + 2\eta$. Define $A_\eta(u)$ as the highest ancestor, including u, with $h(A_\eta(u)) - h(u) \le 2\eta$. This uses the accumulated ancestor-height difference in scalar units, on the reduced equal-level tree with the fixed labels of Eq. 5.

Let $\mathscr{P}_\eta$ consist of two types of leaf-set pairs: $(\mathscr{D}(u), \{k\})$ for every internal node u and leaf $k \notin \mathscr{D}(A_\eta(u))$; and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w satisfying

$$\begin{aligned} \mathscr{D}(w) \cap \mathscr{D}(A_\eta(u)) &= \varnothing, \\ \mathscr{D}(u) \cap \mathscr{D}(A_\eta(w)) &= \varnothing. \end{aligned} \tag{19}$$

For any nonempty leaf set X, use its reference hull $[m_X, M_X]$ and define $\kappa(X,Y) = [\min(M_X - m_Y, M_Y - m_X)]^+/(2L)$. The two certificates are

$$\begin{aligned} C_\eta(T,q) &= \max_{(X,Y) \in \mathscr{P}_\eta} \kappa(X,Y), \\ C_{w,\eta}(T,q,W) &= \max\{E_\partial, \max_{(X,Y) \in \mathscr{P}_\eta} \omega(X,Y)\}. \end{aligned} \tag{20}$$

Theorem 3 (Bounded-Topology Position Bound)**.** *In a profile with $d_{\mathrm{top}} \le \eta$, no leaf outside $\mathscr{D}(A_\eta(u))$ can lie between two leaves of u. Every pair in $\mathscr{P}_\eta$ lies wholly in one block orientation or the other. Consequently,*

$$\mathrm{NROE}(\pi) \ge C_\eta(T,q), \qquad \tau(\pi) \ge C_{w,\eta}(T,q,W), \tag{21}$$

with the second inequality applying to the interval model.

The separation principle excludes interleaving for each eligible pair; the two possible orientations then yield κ and ω. Thus C_η bounds normalized reference-order error, while $C_{w,\eta}$ includes finite-width and boundary costs in coordinate units. Flattening every parent gap of at most 2η and applying Theorems 1 and 2 to the remaining tree gives a second valid certificate. Neither dominates: accumulated gaps can make the chain certificate much stronger (Section 5.2), while flattening also separates pairs of single leaves, which $\mathscr{P}_\eta$ omits. We report the larger of the two.

The allowance concerns the displayed hierarchy; the source field and feature identities stay fixed. A larger feasible set need not improve every feature or every heuristic result. Ranking and acceptance use the exact fixed-order evaluations of Section 4.3, not these bounds.

Scope. The bounds depend only on admissibility (Section 4.1), not on the objective or the solver. Theorem 1 holds for every map that keeps the merge tree exactly, whatever its width rule; Theorem 2 for exact-topology maps in the interval model $\mathscr{L}$; and Theorem 3 for every map whose merge error stays within η. For an exact-topology map, the order error above ν^* is therefore a consequence of its order rule rather than of the data.

5 Experiments

The experiments ask whether the bounds are accurate (Section 5.2), how AnchorMTM compares with existing merge tree maps (Section 5.3), what a merge-level allowance buys (Section 5.4), how the maps read (Section 5.5), and what the choice of reference and the construction cost (Section 5.6).

5.1 Experimental Setup

Data. The benchmark comprises seven sequences and 917 frames (Table 1): reanalysis pressure, satellite radiation, fire emission and air quality on planar domains, two volumetric flow simulations, and the synthetic Ring sequence of TMTM [7,10,14,19,22,23,30,36]. All maps share the processed fields, augmented trees, leaf supports, correspondences and references. References follow Eq. 8 with the estimated motion direction (Eq. 7) on every sequence except Cylinder, where we supply the streamwise axis; on Storms the estimated direction lies 1.1° from west–east. Two further sequences illustrate the maps and enter none of the benchmark aggregates: the 744 hourly Storms fields of Figure 1 and a 90-frame volumetric Vortex sequence.

Compared methods. AnchorMTM runs in Exact Mode ($\varepsilon = 0$) and Budgeted Mode ($\varepsilon = 10\%$ unless stated) with its default configuration Θ_0. We compare it with the native outputs of TMTM [22] and ST-MTM [9]. TMTM uses our port of its official implementation, which matches the official core in all 3127 arc intervals and anchors of the planar sequences. ST-MTM has no public code; our implementation of its description orders leaves by optimal leaf ordering [2], meets the optimality conditions of its quadratic programs (relative KKT residuals below 10^{-6}) and uses one untuned setting on all sequences (Supplement S10). The *projection* places every leaf at its reference and ignores the tree. Native outputs also differ from AnchorMTM in widths and canvas; to compare objectives alone, Figs. 5 and 6 and the Storms case solve two objectives modelled on these methods in our interval model with exact topology and no reference term or cap: temporal alignment (*TMTM-style*) and support-centroid distances with temporal alignment (*ST-MTM-style*); they do not reproduce the original methods. On the planar sequences we also order the extrema along Hilbert and Morton curves. On OLR we also draw a Hovmöller diagram [20] along the same reference, averaging the field over lines perpendicular to it (Supplement S13).

Measures. Position is measured by NROE (Eq. 3), which compares maps with different native scales by their order alone, and, for maps in the interval model, by the pooled mean E_{pool} and the worst E_∞ of $|x_i - q_i|/L$ over all feature–frames, and by NPE, the worst $|x_i - q_i|/L$ of a frame averaged over frames. Topology is measured by NMLE, the rendered d_{top}/Δ_F (Eq. 5). Temporal behaviour is measured by the motion error (ME), the fraction of clear approach or separation events, i.e., changes of the 2D or 3D distance between two matched extrema over two steps, that a map shows in the wrong direction or not at all; by TD, the summed map displacement of matched features [9]; and by the layout-induced jump $|\Delta x_i - \Delta q_i|/L$ of matched features per step. To credit the objective of ST-MTM, we also report scale-normalized stress (SNS) and trustworthiness (TW, $k = 1$) of anchor distances against support-centroid distances with its protocol [9]. TMTM and ST-MTM keep the merge tree exactly, so we measure them against the strict bound $\nu^*(T,q)$. Exact Mode keeps the numerical allowance, so we measure it against $c_0 = C_\eta$ at $\eta = \varepsilon_0\Delta_F$ (Theorem 3), which its order search attains on all 917 frames. The two bounds differ only on Cylinder, where near-tied saddles lower the bound from 12.24% to 8.01% at a merge error below $10^{-7}\Delta_F$ (Supplement S16).

Table 1: Benchmark sequences. Leaves: retained extrema per frame.

Sequence	Scalar field	Frames	Leaves	Tree
Storms	ERA5 mean sea-level pressure, 2D	118	1–8	join
Wildfires	log(1+FRP), 2D	30	6–11	split
OLR	outgoing longwave radiation, 2D	60	2–8	join
$PM_{2.5}$	$PM_{2.5}$ concentration, 2D	366	1–7	split
Ring	synthetic ring intensity, 2D	40	1–7	split
Tangaroa	velocity magnitude, 3D	201	18–93	join
Cylinder	Q criterion, 3D	102	4–75	split

5.2 The Bounds Are Attained in Controlled Trees and Informative on Real Ones

Each bound isolates one cause of displacement. To separate the causes, we fix two sibling pairs $X = \{A,B\}$ and $Y = \{C,D\}$ that merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5-d, 0.8-d)$ on a unit canvas, and vary the signed overlap d from -0.12 to 0.30. Across 43 overlaps and four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax LP. Topology alone forces exactly $\nu^* = \max(0,d)/2$. With widths $W = 0.06$, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0,(d+0.04)/2)$ (Fig. 3a; Fig. 2c): 12% and 14% at $d = 0.24$. A merge allowance removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps tested (Fig. 3b). These are controlled instances, not a general tightness claim.

On real trees, the bounds remain informative. The interval cost τ of the selected order meets the width bound on most planar frames and on the worst Cylinder frame at every tested budget, but not uniformly: on Tangaroa the worst frame meets it only at $\varepsilon = 2\%$, and on Wildfires at $\varepsilon > 0$ and on one OLR frame at $\varepsilon = 10\%$, a worst-frame gap remains. Accumulating ancestor gaps keeps the bounded-topology bound useful on large trees: on Tangaroa at $\varepsilon = 5\%$, flattening every gap below 2η certifies only 0.38% on average, whereas Theorem 3 with widths certifies 7.40% against an attained 8.10%. A gap of similar size remains at $\varepsilon = 0$; it does not tell slack in the bound from a suboptimal search. Per frame, the certificates thus tell an analyst how much of a map's displacement no layout in the model could avoid.

5.3 Comparison with Existing Merge Tree Maps

Exact Mode nearly attains the exact-topology bound, and most of the order error of TMTM and ST-MTM lies above it. Exact Mode lies within 0.03 percentage points of the bound on every planar sequence, with zero merge error, and attains c_0 on Cylinder (Fig. 4). On Tangaroa, the width-aware search accepts a slightly higher NROE (9.67% against 9.24%) to lower the worst actual displacement (Section 5.6). TMTM and ST-MTM also keep the merge tree, so Theorem 1 splits their order error into the part ν^* that the hierarchy forces and an excess that their order rule chooses and that another order of the same tree would avoid. The excess accounts for 54–95% of TMTM's order error and 31–95% of ST-MTM's; the smallest shares are on the volumetric sequences, where the hierarchy forces more. On Storms, for example, TMTM's 19.2% contains 3.8% that the hierarchy forces and ST-MTM's 16.3% contains 12.5% that it does not. Space-filling curves, which order the domain rather than the tree, are no remedy: Hilbert and Morton orders of the extrema, averaged over the eight curve orientations, have NROE of 8.3–23.4% on the planar sequences, close to TMTM and ST-MTM,

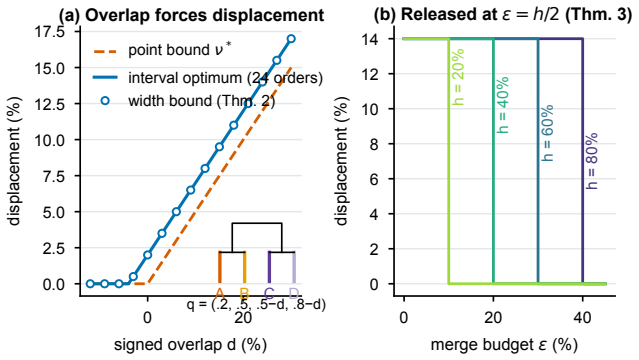


Figure 3: Controlled four-leaf trees (inset: sibling pairs and references). (a) Increasing sibling-range overlap d forces displacement; the width bound of Theorem 2 (circles) equals the optimum over all 24 orders. (b) At $d = 24\%$, the 14% displacement disappears exactly when the merge budget reaches half the parent gap h, for all four gaps, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

and even with optimal barriers they change merge levels by up to 19–39% of the scalar range (Supplement S10).

The anchored axis carries motion. The motion term rewards displayed motion that follows projected motion, and Exact Mode makes fewer motion errors than TMTM and ST-MTM on all seven sequences, for example 5.4% against 21.4% on OLR and 52.2% against 70.0% on Tangaroa. The projection, which follows the reference order exactly, is lower still on most sequences, but only by breaking the tree: it changes merge levels by up to 58.66% of the scalar range on Storms, 52.58% on Wildfires, 33.56% on OLR and 31.02% on Tangaroa, where no frame keeps its tree. Budgeted Mode closes part of this gap within its allowance (Section 5.4); no rendered AnchorMTM map exceeded its declared allowance beyond numerical tolerance.

The price of a spatial axis is stated, not hidden. Anchoring trades two-dimensional distance preservation for position along one direction. ST-MTM, whose objective is pairwise distance, has the best TW on all seven sequences and the best SNS on five; the space-filling curves are worse than ST-MTM in both on every planar sequence. Where widths dominate, the trade-off can reverse: on Cylinder, Budgeted Mode reaches an SNS of 0.041 against ST-MTM's 0.135.

5.4 One Control Traces the Position–Topology Trade-off

One control spans the trade-off from exact topology to the reference order, and where the curve bends depends on the source of the conflict. Figure 5 traces each sequence from Exact Mode to the projection; every point respects its allowance. Where conflicts come from packing, small budgets suffice: on Cylinder, $\varepsilon = 1\%$ lowers the worst displacement from 27.00% to 5.27%, below the projection's 5.73%, because sorting by reference ignores widths; on Wildfires, 1% lowers E_∞ from 20.48% to 8.71%; on $PM_{2.5}$, 5% already reaches the projection. Where conflicts come from strong saddles, position is expensive: on Storms, the worst displacement stays above 18% up to $\varepsilon = 20\%$, and on OLR one frame keeps $E_\infty = 17.14\%$ up to 10%. Budgets are spent where they are needed: on Storms at 10%, the mean merge error is 0.90% while the worst frame uses 8.51%. For comparison, the TMTM-style and ST-MTM-style objectives, solved in the same interval model with exact topology, leave worst displacements of 48–76% and 61–99% and mean displacements of 17–27% and 14–43% on the planar sequences (Fig. 5, grey boxes; Supplement S11), whereas on these sequences no point of the AnchorMTM curves exceeds 26.2% and 4.2%.

Aggregate gains are not pointwise gains. On the Vortex diagnostic, raising ε from 20% to 30% more than halves the pooled error (8.79% to 4.27%), while 30.91% of 12,446 feature–frame observations worsen, 24.72% by at least 0.5 percentage points of L.

5.5 Reading the Maps

OLR: features on their references, with their values and nesting. Figure 6 shows the same 60 days in a Hovmöller diagram and in four merge tree maps. Only the anchored maps keep the low-radiation basins on their references: the TMTM-style and ST-MTM-style objectives leave mean displacements of 24.4% and 42.7% of the axis, Exact Mode 1.0%. The Hovmöller diagram places every sample exactly but averages across the reference. Of the 264 minima, its profile shows a separate dip within 2% of the reference for 59% (77% within 5%), and these dips are on average $32\,\mathrm{W\,m^{-2}}$ shallower than the minima; minima at similar reference positions but different latitudes merge into one dip. By the same test, Exact Mode has a dip within 2% (5%) for 94% (97%) of the minima, and 93% are shown at their exact values within 2%; the others are displaced farther or swapped with a neighbouring minimum. Their nesting is kept. The Hovmöller diagram is smoother in time: a minimum appears or vanishes between every pair of consecutive days, and a map that keeps each day's tree changes with it (mean column change 9.0% of the scalar range, against 4.1%). The two displays answer different questions and can be read side by side.

Storms: an exact map that reads as geography. On the 118-frame Storms benchmark, the minima are low-pressure systems crossing the North Atlantic, and the reference runs west–east. Measured by NPE, the TMTM-style and ST-MTM-style objectives place features 37.3% and 50.7% of the canvas from their references within the same layout model, and Exact Mode 5.1% (median per frame 0.8%, peaks of 26.2% in a few frames). A 10% budget lowers NPE to 2.7%; it is spent in 31 of 118 frames, at most 8.51% and 0.90% on average, and the 89 barriers whose merge level changed by more than 0.1% of the scalar range are marked on the map (Supplement S12). Figure 1 applies the construction at $\varepsilon = 5\%$ to 744 hourly fields of December 1999 with a fixed projected direction. Its snapshots follow one matched minimum, a strong storm over Scandinavia, from 16 to 19 December, whose displayed positions form one continuous track through the outlined window; the track is a computational correspondence of leaf features, not by itself evidence of physical storm genesis or dissipation. Supplement S15 follows one tracked feature per dataset on five sequences through six budgets from 0 to 30%, showing feature by feature what a larger budget changes.

5.6 Reference Choice, Ordering and Cost

Because the bounds need no layout, they inform the choice of reference before a map is drawn. In 3D, a projection discards two coordinates, and the unavoidable order error depends on the direction. We evaluated the bound c_0 for 400 sampled directions on both volumetric sequences, with references built as in Eq. 8 (maps over the hemisphere are in Supplement S14). Over these samples its mean ranges from 4.70% to 10.14% on Tangaroa and from 5.49% to 28.15% on Cylinder. On Tangaroa, the estimated motion direction is among the most conflicting: only 4 of the 400 sampled directions have a higher bound, and the x axis would lower it from 9.24% to 6.09%. On Cylinder, the streamwise axis lies near the median (8.01%). In every key direction, Exact Mode stays within 0.6 percentage points of the bound, except on the Tangaroa z axis (2.6 points), where large widths dominate, while TMTM and ST-MTM stay far above it.

Width-aware ordering. An order that is optimal for points need not be optimal for intervals when features are large. On Tan-

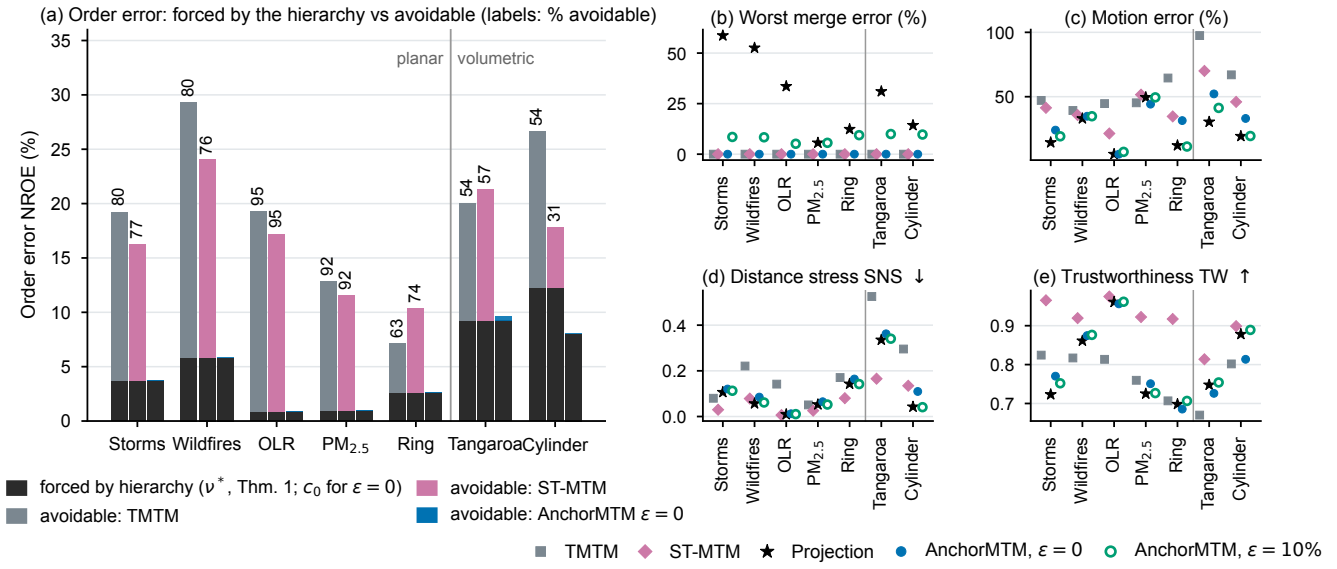


Figure 4: Native outputs of TMTM and ST-MTM, the projection and AnchorMTM on seven sequences (left of the grey lines: planar; right: volumetric). (a) Order error NROE split by Theorem 1 into the part forced by the hierarchy (black: ν^* for TMTM and ST-MTM, c_0 for Exact Mode; they differ only on Cylinder, Section 5.1) and the avoidable excess of each map; labels give the avoidable share in percent. (b) Worst rendered merge error: exact-topology maps have none, the projection up to 59%. (c) Motion error. (d, e) The distance measures of ST-MTM. All values are listed in Supplement S10.

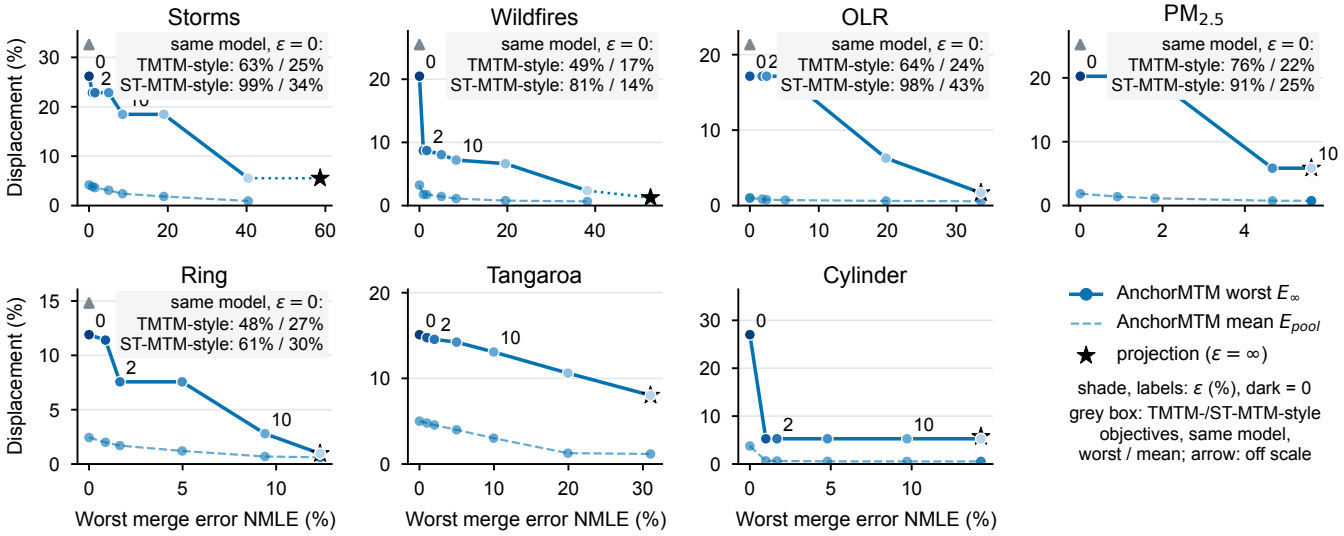


Figure 5: Position–topology trade-off traced by the single control ε. Each point is one map of all frames at budget ε (shade and labels, in percent), placed by its rendered worst merge error and its worst (E_∞, solid) and mean (E_{pool}, dashed) displacement. Stars: projection ($\varepsilon = \infty$). Grey boxes: worst and mean displacement of the TMTM-style and ST-MTM-style objectives (Section 5.1), above the plotted range (arrows).

garoa, ordering by NROE alone, i.e., the point-model budget order, leaves a worst displacement of 20.69% at $\varepsilon = 0$. Ranking candidates by the exact interval cost τ, as Algorithm 1 does, lowers it to 15.10% at $\varepsilon = 0$ and to 14.55% at $\varepsilon = 2\%$, where the worst frame's order meets the width bound, at a mean temporal jump of 2.82% against 2.73%. On Cylinder, $\varepsilon \geq 2\%$ lowers E_∞ from 5.73% with the point-model order to 5.27%, again at the bound.

Search coverage and runtime. On all 637 frames with at most eight leaves, the point-model order search attains the exhaustive NROE optimum at nine budgets from 0 to 100%. The bounds cost $O(n \log n)$ to $O(n^2 \log n)$ and δ^* costs $O(n^3)$ per order. On an Apple M2 Pro (single Python process; NumPy 2.0, SciPy 1.13), all bounds take at most 0.6 ms per planar frame (1–11 leaves) and 5.5 ms per volumetric frame (up to 93 leaves), the ordering at most 3.7 ms and 103 ms, and the capped QP (SciPy SLSQP) 2.3–9.0 ms and 0.25–0.50 s. All 917 frames take 94 s at $\varepsilon = 0$ and at most 137 s with a budget.

6 Discussion and Limitations

What the bounds are for. The bounds need neither a layout nor a solver. Before a map is drawn, they compare candidate references in milliseconds (Section 5.6); after it is drawn, they sep-

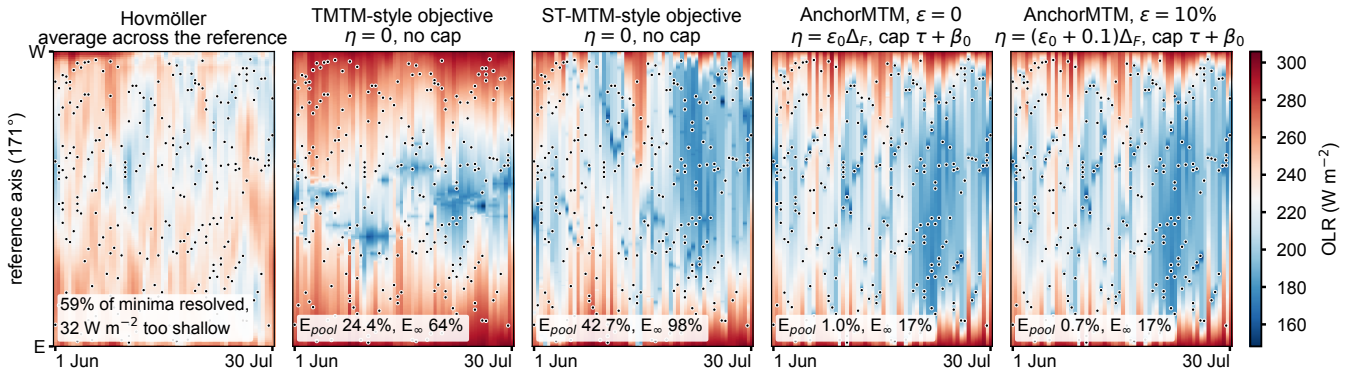


Figure 6: Sixty days of OLR (June–July 2024, tropical Indian Ocean and western Pacific) on one reference axis, the estimated direction 171°, running from east to west. Left: Hovmöller diagram along the reference. Right: four merge tree maps, labelled by their allowance and cap. Black dots: reference positions of the minima. All panels share one colour scale; labels give the share of minima resolved by the Hovmöller profile within 2% of their reference and the mean depth error of those dips, or the displacement of the merge tree maps.

arate the displacement that the hierarchy forces from the part another layout could remove, for AnchorMTM and for any other map that meets their constraints. Because midpoint ordering attains ν^*, any order error above it is avoidable without changing the tree; the width and bounded-topology bounds need not be attained, so the excess above them estimates from above what a better layout could remove.

Stable is not faithful. Temporal stability measures reward maps that do not move. On Storms, the projection, which follows the references, accumulates a TD of 23.1 canvas units; TMTM and ST-MTM move their features by 10.1 and 7.2 and show 47% and 41% of clear approach and separation events wrongly, against 14% for the projection and 24% for Exact Mode. AnchorMTM's movement mostly follows the reference: on OLR, the layout-induced jump averages 1.0% of the axis per step against an anchor motion of 5.9%. Merge tree maps also change where the tree changes: on Storms and Ring, the columns of every map change two to three and a half times more at steps where a tracked minimum appears or vanishes. On volumetric data, layout jumps are as large as the anchor motion (2.8% per step on Tangaroa), a cost that a temporal term in the order selection could reduce.

What a budget changes. A budget never removes a feature, changes an extremum value or moves the data. In the continuous model it changes merge levels by at most $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, and every rendered map we measured stayed within this allowance. Where the labels cover all leaves, as we verified on 200 rendered columns of Ring and OLR, the displayed tree and its persistence diagram stay within the measured merge error of the source in interleaving and bottleneck distance. Persistence simplification at threshold 2η instead deletes the features below it [12, 33]. Where a subtree is split, its band is no longer contiguous and "one band, one component" fails locally, which is why such barriers are marked.

Using AnchorMTM. The results suggest a simple procedure. First, compare candidate references by ν^*, which takes milliseconds per frame, and choose between the estimated motion direction and a physically meaningful one. Second, start in Exact Mode, whose position error is close to the certified bound on planar data and whose merge tree is exact. Third, sweep ε, choose the smallest budget at which the trade-off curve drops, and inspect the marked barriers and per-feature residuals before accepting it. For reading propagation speed, a Hovmöller diagram along the same reference complements the map.

Limitations. One direction cannot encode the other coordinates: features with the same projection compete for the same place, and the estimated direction, which follows the dominant motion, need not be the least conflicting one. It is estimated offline from all matches; online use requires a supplied direction or one fitted to an earlier interval. The construction inherits the correspondence: missed or spurious matches change the estimated direction and the motion term. The order search is a finite heuristic; its point-model stage matched exhaustive enumeration in NROE on all frames with up to eight leaves, and on larger trees its excess is limited only by the gap to the bound. Continuous feasibility does not certify the rendered raster: resampling, strict-extremum clearance and pixel rounding can change the achieved merge error, so we report rendered errors separately from δ^*. All results use the default configuration Θ_0; we did not study its sensitivity. Temporal behaviour and relative geometry enter as objectives and are measured, not certified. Finally, the motion error is a geometric proxy that assumes known identities; whether analysts read position and nesting together, and whether disclosure marks prevent misreading [21, 25], requires a user study.

7 Conclusion

AnchorMTM gives temporal merge tree maps a spatial meaning: a reference turns the display axis into a spatial coordinate, one control decides how much topology may change to follow it, and three certified bounds state what exact topology, finite widths and a bounded topological error make unavoidable. The bounds depend only on the layout constraints and so also measure other maps: on seven planar and volumetric sequences, 31–95% of the order error of TMTM and ST-MTM lies above the exact-topology bound, and Exact Mode removes nearly all of it on planar data without changing the merge tree. Next steps are scalable order search for large trees, references beyond a single direction, certificates for temporal behaviour, and a user study.

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